

THE LOCAL GROUP OF GALAXIES

The Milky Way is one of a cluster of galaxies, called the Local Group, that occupies a region 10 million light-years across. It contains more than 50 known galaxies, only one of which—the Andromeda Galaxy—of similar size to the Milky Way. Most others are small (dwarf) galaxies.

THE LOCAL SUPERCLUSTER

The Local Group of galaxies, together with some nearby galaxy clusters, such as the giant Virgo Cluster, is contained within a vast structure called the Virgo Supercluster. It is 110 million light-years across and (if dwarf galaxies are included) contains tens of thousands of galaxies.

DISTANT GALAXY CLUSTER

The vast galaxy cluster Abell 2218 (left) is visible from Earth even though it is more than 2 billion light-years away.

LARGE-SCALE STRUCTURE

Galaxy superclusters clump into knots or extend as filaments that can be billions of light-years long, with large voids separating them. However, at the largest scale, the density of galaxies, and thus all visible matter, in the universe is very uniform.

Ursa Minor
dwarf galaxy

Milky
Way

250,000
light-years

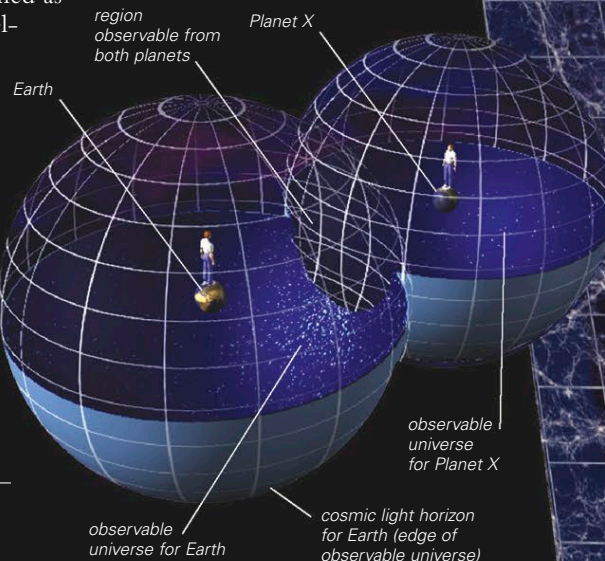
Leo A

10 million
light-years

100 million
light-years

THE OBSERVABLE UNIVERSE

Although the universe has no edges and may be infinite, the part of it that scientists have knowledge of is bounded and finite. Called the observable universe, it is the spherical region around Earth from which light has had time to reach us since the universe began. The boundary that separates this region from the rest of the universe is called the cosmic light horizon. Light reaching Earth from an object very close to this horizon must have been traveling for most of the age of the universe—that is, approximately 13.8 billion years. This light must have traveled a distance of around 13.8 billion light-years to reach Earth. Such a distance can be defined as a “lookback” or “light-travel-time” distance between Earth and the distant object. However, the true distance is much greater, because since the light arriving at Earth left the object, the object has been carried farther away by the universe’s expansion (see p.45).

**OVERLAPPING OBSERVABLE UNIVERSES**

Earth and Planet X—an imaginary planet with intelligent life, located tens of billions of light-years away—would have different observable universes. These may overlap, as shown here, or they may not.

FROM HOME PLANET TO SUPERCLUSTERS

The universe has a hierarchy of structures. Earth is part of the solar system, nested in the Milky Way, which in turn is part of the Local Group. The Local Group is just part of one of millions of galaxy superclusters that extend in sheets and filaments throughout the observable universe.